8 Fig. 8.1 shows the cross-section of a straight driveway 4 m wide made from tarmac.

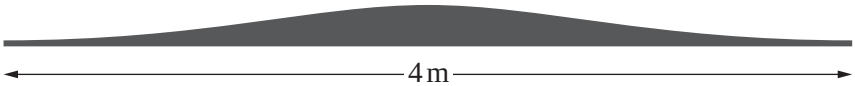


Fig. 8.1

The height h m of the cross-section at a displacement x m from the middle is modelled by $h = \dfrac{0.2}{1+x^2}$ for $-2 \leqslant x \leqslant 2$.

A lower bound of $0.3615\,\text{m}^2$ is found for the area of the cross-section using rectangles as shown in Fig. 8.2.

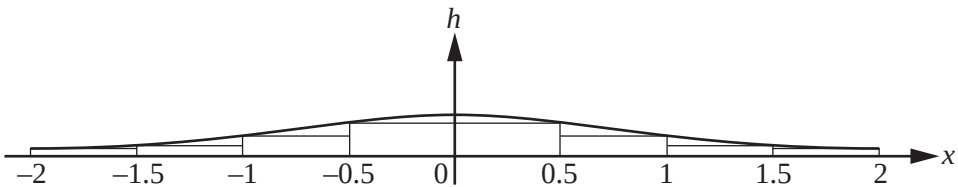


Fig. 8.2

(a) Use a similar method to find an upper bound for the area of the cross-section. **[3]**

(b) Use the trapezium rule with 4 strips to estimate $\displaystyle\int_0^2 \frac{0.2}{1+x^2}\,\mathrm{d}x$. **[2]**

(c) The driveway is 10 m long. Use your answer in part **(b)** to find an estimate of the volume of tarmac needed to make the driveway. **[2]**

8(a)

8(b)

8(c)